

Linear systems and elimination

Exponential smoothed tail bounds for partial pivoting

Difficulty: extreme **Importance:** broadly interesting

Status: open.

Context and notation

All elimination in this problem is in exact arithmetic. Write $\|A\|_{\max} = \max_{ij} |a_{ij}|$. A pivoting path creates successive active Schur complements $S_1 = A, S_2, \dots, S_n$, with row/column permutations as appropriate. Its element-growth factor is

$$\rho(A) = \frac{\max_{1 \leq j \leq n} \|S_j\|_{\max}}{\|A\|_{\max}}.$$

Partial pivoting chooses a largest-magnitude entry in the active first column; complete pivoting chooses one anywhere in the active matrix. If ties occur, a universal statement includes every admissible tie choice; a supremum includes all admissible paths. These conventions remove implementation-dependent ambiguity. Matrices are nonsingular unless otherwise stated.

Problem statement

Do universal constants $c_1, c_2 > 0$ exist such that, for every $n \geq 1$, deterministic $\bar{A} \in \mathbb{R}^{n \times n}$ with $\|\bar{A}\|_2 \leq 1$, $0 < \sigma \leq 1$, and $x \geq 1$, the matrix $A = \bar{A} + \sigma G$, with independent standard normal entries in G , satisfies

$$\Pr\{\rho_{\text{PP}}(A) > x(n/\sigma)^{c_1}\} \leq 2^{-c_2 x}?$$

This is a uniform smoothed-analysis question: the deterministic center may itself be a worst-case input. Almost surely the perturbed matrix is nonsingular and the pivot choices have no ties. Numerical evidence or estimates only at $\bar{A} = 0$ do not settle the quantifier over deterministic centers. The tail estimate would quantify the rarity of substantial growth after small Gaussian input perturbations.

References

Spielman and Teng, *Smoothed Analysis: An Attempt to Explain the Behavior of Algorithms in Practice*, §P6, Conjecture 16, p. 52 of the author PDF. Huang and Tikhomirov, *Average-case analysis of the Gaussian elimination with partial pivoting*, PTRF 189 (2024), introduction.

Status check

Searches for *Exponential Stability of GEPP solved* and *Gaussian elimination smoothed analysis 2025 2026* found the average-case theorem, not this exponential tail bound. The 2026 Peca-Medlin butterfly paper also distinguishes average-case results from the still-unavailable full smoothed analysis. Randomizing the pivot rule is a different model.